A Hybrid Framework for Efficient and Robust Deepfake Detection

PROJECT REPORT

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In partial fulfillment of the requirements for the award of the degree of

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Under the Guidance of

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Certificate

This is to certify that the project report entitled **"A Hybride Framework For Efficient And Robust Deepfake Detection"** is a bonafide work carried out by **Navanith Krishna R (1BM22CS172), Navya Billalar (1BM22CS175), Nikhilesh C (1BM22CS181), Pooja Gaikwad (1BM22CS194)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Computer Science and Engineering**, under my supervision during the academic year 2025–2026.

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Abstract

This thesis presents research on enhancing the reliability and robustness of deepfake detection systems through a lightweight and intelligent hybrid architecture. The proposed framework integrates Convolutional Neural Networks (CNNs) with Transformer encoder blocks to process both spatial (RGB) and frequency-domain features, enabling accurate identification of manipulated facial content.

To address limitations of existing models such as overfitting to dataset-specific artifacts and reliance on large labeled datasets the system employs self-supervised adversarial fake generation and is trained under various real-world distortions, including compression, blur, and noise. The resulting AI-powered system is efficient, scalable across diverse datasets, and suitable for deployment on resource-constrained devices. It provides a reliable tool for digital media verification and enhances resilience against advanced tampering techniques.

DECLARATION

We, hereby declare that the Major Project Phase-2 work entitled “A Hybride Framework For Efficient And Robust Deepfake Detection” is a bonafide work and has been carried out by us under the guidance of ANUSHA S, Associate Professor, Department of Computer Science and Engineering, B.M.S. College of Engineering, Bengaluru, in partial fulfillment of the requirements of the degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi. We further declare that, to the best of our knowledge and belief, this project has not been submitted either in part or in full to any other university for the award of any degree.

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Acknowledgment

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List of Acronyms

Below is a list of acronyms and abbreviations used throughout this report for clarity and consistency.

AI Artificial Intelligence

CNN Convolutional Neural Network

FFT Fast Fourier Transform

GAN Generative Adversarial Network

GPU Graphics Processing Unit

ML Machine Learning

MTCNN Multi-task Cascaded Convolutional Networks

RGB Red Green Blue (Color Space)

SDG Sustainable Development Goals

TPU Tensor Processing Unit

UDA Unsupervised Domain Adaptation

ViT Vision Transformer

DFDC DeepFake Detection Challenge

FF++ FaceForensics++

SFF Specific Forgery Features

CFF Common Forgery Features

API Application Programming Interface

SSD Solid State Drive

SOTA State of the Art

ROC-AUC Receiver Operating Characteristic – Area Under Curve

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Chapter 1 Introduction

1.1 Background

The integrity and authenticity of digital media have been seriously threatened by the swift development of generative technologies like diffusion models and Generative Adversarial Networks (GANs). Deepfakes, which are extremely realistic synthetic audio, video, and image content, are made possible by these techniques. These innovations have the potential to benefit the creative and entertainment sectors, but they have also turned into potent instruments for bad actors, which can have detrimental effects like identity theft, privacy violations, political disinformation, and representational damage. By making high-quality forgeries accessible to the general public, the widespread availability of user-friendly deepfake generation tools has increased these risks and raised the possibility of misuse in real-world situations.

Conventional deepfake detection methods rely on pixel-level inconsistencies or forensic cues that were made by hand. However, these methods have a lot of problems. For example, they are easy to trick with new manipulation techniques, they are sensitive to noise, occlusions, and compression artifacts, and they need large, well-annotated datasets to train quickly. When tested on cross-dataset or real-world samples, even sophisticated deep learning-based detectors, such as Convolutional Neural Networks (CNNs) and Transformer architectures, frequently perform worse—a problem commonly referred to as the domain gap or poor generalization. Additionally, the deployment of such high-capacity models on mobile or resource-constrained platforms is limited by their computational complexity, and ongoing advancements in generative model design frequently outpace corresponding detection mechanisms.

As a result, there is an urgent need for deepfake detection frameworks that are lightweight, strong, and can be used in many different situations. These frameworks should be able to adapt to new manipulation methods, stay accurate even when there are different kinds of distortions, and be easy to use for media authentication, misinformation mitigation, and public safety in today's digital world.

1.2 Problem Statement

A detection solution that is practical, efficient and applicable to the real world is a necessary time-sensitive requirement. For example, one stipulation that typically is limiting is that large, expensive, labeled datasets are a cumulative requirement over time; however, as deepfake generation methods are temporally based, our datasets get outdated quicker than we like and do not successfully generalize over different types of attack. But an optimal framework would detect forgeries from unseen attacks, advanced generative methods and actual media of real-world diversities, encompassing compression, blur and noise. This means that a detection solution on which we could depend that would not require shifting or adjusting would depend on consistent variables. We need the ability to adjust to current times and a detection solution that is always responsive to situational changes.

A key factor that makes detection successful is the use of spatial (RGB/pixel-based) and frequency-based features. Spatial features recognize tiny differences between pixels while frequency based features allows for recognition of inconsistencies in texture or spectral patterns that aren't necessarily visible on the RGB spectrum. Multispectral or single-domain technologies can fall victim to advanced forgeries that are computer generated to avoid any signals of detection. Therefore, multispectral information provides a more comprehensive understanding which, ultimately, is less likely to be duped by advanced forgeries and more effective in the real world.

A third challenge concerns computational efficiency and scalability. Most complex detectors rely on high capacity deep learning architectures that are lacking in inference speed for real-time application or integration onto mobile/embedded systems. In addition, studies show that these networks experience a 40–50 percentage reduction in performance on real-world datasets (domain gap). Therefore, this project aims to implement a hybrid deepfake detection framework that prioritizes spatial and spectral detection, requires minimal labeled data via self-supervised learning, is low power for real-time operation, yet is generally effective and trusted in diverse and unpredictable real-world settings.

1.3 Objectives

The objectives of this project are carefully formulated to address the complex challenges associated with deepfake detection in real-world environments.

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- **Hybrid Model Design:** The approach for developing a hybrid deepfake detection model consists of two types of architecture. For this purpose, we will use a dual-branch model that combines a convolutional neural network (CNN) with a transformer encoder. These two branches will be used to process two different levels of information: The "fine-grain" level of detail captured through the original RGB images processed by the CNN and the larger context level generated through the frequency domain (via wavelet or Fourier transformations) by the transformer encoder. This method is expected to provide better detection of small and/ or subtle manipulation artifacts than would be achieved using only one of the two architectures.

- **Self-Supervised Learning Integration:**Self-supervised learning methods allow for the creation of adversarial samples that imitate numerous different types of manipulation with less dependency upon human provided labels or annotations. Consequently, self-supervised learning permits the application of this type of method to novel and emerging kinds of forgery by eliminating the need to have large labelled datasets.

- **Computational Efficiency and Deployment:**The Deployment and Efficiency project will concentrate on optimising Frameworks for Lightweight Designs and Efficient Inference, while remaining compatible with platforms that are resource constrained. An appropriate trade-off between Optimisation Factors (i.e., Accuracy, Speed and Scalability) is anticipated at all times, relative to compatibility with real-world deployment.

- **Robustness Evaluation:** Assess the likelihood of success for the suggested approach by rigorously evaluating the approach's capacity to endure the frequent errors that occur when applying digital image processing algorithms to real-world images, including distortions from compression, blur, and noise. This evaluation will give you assurance that the suggested method will remain functional in whatever unpredictable environments may arise.

The framework for designing a hybrid deepfake detection system that meets both academic research objectives and the operational needs of the digital media industry. Efficient, adaptable, and effective hybrid deepfake detection systems are an important part of any digital media environment and help ensure that the development and use of such systems continues to be a focus for ongoing academic research.

Chapter 2 Literature Survey

2.1 Overview

Numerous studies have been done to investigate various methods of deepfake detection using spatial information, frequency transformations, and combinations of advanced network models. While these methods demonstrate significant success within controlled datasets, they demonstrate considerably diminished performance when evaluated against previously unseen manipulative data and when evaluated in realistic settings. This literature review provides the central contributions and deficiencies from some of the most prominent studies in this area and establishes a basis for future research into this area.

2.2 Scope and Boundaries of Literature Review

This literature review looks broadly at recent progress in **deepfake detection research**, with an emphasis on how different methodological directions attempt to improve reliability and generalization in real-world settings. In doing so, the discussion spans a fairly wide range of approaches, including **spatial (RGB/pixel-level) analysis**, **frequency-domain techniques**, and several forms of **hybrid feature fusion**. A particular focus is placed on works that build on **CNN-based designs** and emerging **Transformer architectures**. Along the way, I also note techniques that support these models—such as **domain adaptation**, various flavors of **self-supervised or adversarial learning**, and attempts at **multi-modal feature integration**—since these appear frequently in more recent studies.

The review is intentionally scoped around **image-based facial forgery detection**, especially methods that aim to maintain **cross-dataset robustness** and stay reliable under **real-world distortions** like heavy compression, blur, or sensor noise. I also highlight research tackling issues such as generalization failure, the push toward lightweight or deployable models, and ongoing attempts to bring some degree of interpretability or explainability into deepfake detection systems—an area that continues to evolve somewhat unevenly.

To keep the review relevant and focused on current trends, the literature surveyed spans approximately **2019–2025**, covering material from **peer-reviewed journals**, major venues such as **IEEE** and **ACM** conferences, and a selection of widely cited open-access publications. Some areas are intentionally set aside—for example, **medical deepfake detection**, **audio-only forensic tools**, and methods based primarily on **digital watermarking**. These are referenced only when needed for broader context, as they fall outside the central objective of this project: developing a **practical, robust, and generalizable facial deepfake detection framework** that can hold up under everyday deployment conditions.

2.3 Flow of Literature Review in the Form of a Diagram

The logical progression of the literature review for this project is depicted in the diagram below. It begins by determining what motivated research into deepfake detection, follows through with an overview of all the different types of methods that have been used so far, includes a summary of important studies and what they added to our understanding of deepfake detection, identifies gaps in the current research and finally highlights how the proposed hybrid framework fits into this picture by providing solutions to the gaps identified.

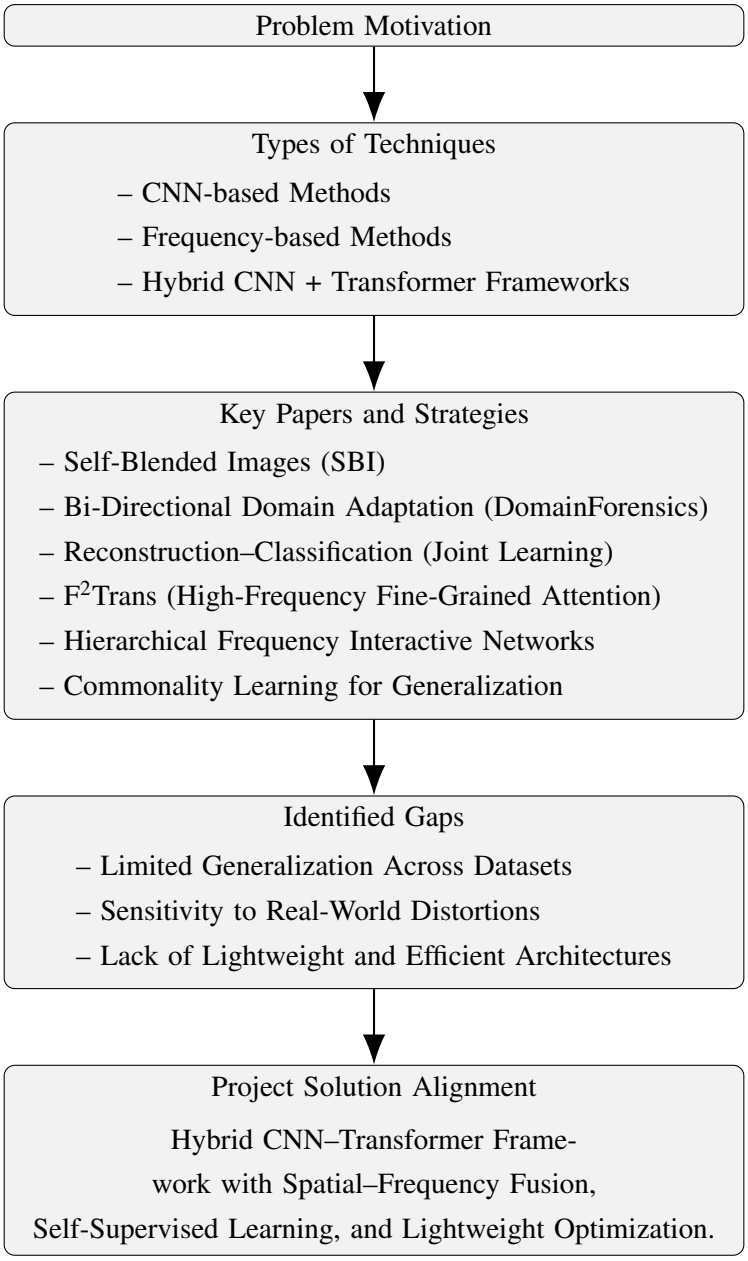

Figure 2.1: Flow of the Literature Review for Deepfake Detection

2.4 Literature Review Table

In the table below, we provide a brief overview of the most recent research into Deepfake Detection methods and summarises the research methodologies, datasets, key contributions, and limitations of each project. This overview allows us to identify current trends, as well as research gaps, challenges that will shape the motivation and focus of our current project.

No.	Paper Title	Year	Approach / Technique	Dataset(s)	Main Contribution	Limitations
1	Detecting Deepfakes with Self-Blended Images	2022	Self-supervised blending	FF++, DFD	Enhanced cross-dataset generalization (+11% AUC)	Some visual artifacts
2	DomainForensics: Bi-Directional Domain Adaptation	2024	Cross-domain adaptation	WildDeepfake	Strong domain transfer; improved robustness to unseen manipulations	Requires dual-domain training
3	End-to-End Reconstruction–Classification Learning	2022	Multi-task: reconstruction + classification	DFDC	Achieves SOTA on DFDC; stable generalization	High computational cost
4	F^2Trans: Fine-Grained Transformer for Forgery Detection	2023	Hybrid transformer (frequency-domain)	Celeb-DF	Detects high-frequency artifacts effectively	Large model size
5	Hierarchical Frequency-Assisted Interactive Networks	2022	Frequency-assisted local–global fusion	FF++, Celeb-DF	Strong performance under noisy conditions	Complex tuning
6	Improving Generalization via Commonality Learning	2022	Domain generalization via feature commonality	DFDC, Celeb-DF	Enhances transferability across datasets	May miss unseen attacks

No.	Paper Title	Year	Approach / Technique	Dataset(s)	Main Contribution	Limitations
7	Systematic Literature Review on Deepfake Detection Techniques	2023	Review (CNN, DNN, LSTM)	FF++, DFD, Celeb-DF	Evaluates 48 models with metrics	Excludes video-only methods
8	Deepfake Detection using Deep Learning Methods: A Systematic Review	2022	Deep learning survey (CNN, RNN)	FF++, DFD, Celeb-DF	Categorizes SOTA detection methods	Limited to 2018–2022 works
9	A Systematic Review of Deepfake Detection and Generation	2024	Multi-modal (audio, video, image)	DFD, DFDC, FaceForen-sics	Covers both detection and generation challenges	Focused more on audio/video
10	Deepfake Detection: A Systematic Literature Review	2023	Classical ML, statistical, blockchain-based	Various datasets	Defines taxonomy and compares techniques	Lacks deep learning focus
11	Deepfake Definitions, Metrics, and Datasets	2024	Benchmark analysis and definitions	FF++, DFD	Summarizes datasets and metrics	No new detection model proposed
12	A Review of Deepfake Techniques: Architecture, Detection, and Challenges	2023	Comprehensive literature review	Multiple datasets	Discusses detection trends and challenges	No experimental validation
13	Advances in Face Forgery Detection: Deep Learning Approaches	2021	CNN / Ensemble / Siamese networks	FaceForensic	Studies ensemble models under low-data settings	Minimal frequency-based analysis

No.	Paper Title	Year	Approach / Technique	Dataset(s)	Main Contribution	Limitations
14	Robustness of Deepfake Detectors under Distortions	2023	Robustness testing (blur/noise)	DFD, Celeb-DF, KoDF	Evaluates models under various distortions	Focuses only on robustness
15	Recent Trends in Deepfake Detection: Challenges and Opportunities	2021	Trend and roadmap analysis	Various datasets	Identifies key research challenges	No implementation insights

Table 2.1: Summary of Deepfake Detection studies .

2.5 Summarize Literature Review

The literature review reveals a clear progression from conventional CNN-based deepfake detectors to more advanced hybrid frameworks that integrate spatial, frequency, and transformer-based attention mechanisms. Pioneering works such as Self-Blended Images and DomainForensics highlight the significance of dataset augmentation and domain adaptation in enhancing detection robustness across varied manipulation types.

Recent research has further explored multi-task learning approaches—such as End-to-End Reconstruction-Classification—and fine-grained spectral analysis methods like F2Trans and Hierarchical Frequency Networks, which have established new benchmarks in detection accuracy and resilience under real-world conditions.

While there has been a lot of progress in developing techniques for training computer vision models to improve their generalization, many studies still struggle with generalizing well to previously unseen data manipulations or image degradation. Research related to domain adaptation, self-supervised learning and model deployment can contribute to better understanding how to create models that work well in a variety of settings and enable these methods to scale and be used in real-world applications.

In general, this evolving body of work directly influences the proposed hybrid, frequency- attentive, and self-supervised deepfake detection framework, which aims to integrate the strengths of prior methodologies while overcoming their limitations in robustness, efficiency, and adaptability.

Chapter 3 Software and Hardware Requirement Specification

A Four-Stage DeepFake Detection System uses Artificial Intelligence techniques to identify whether Media Content is Fake or Authentic. The system employs a combination of Preprocessing, Feature Extraction, Modelling/Inference and Reporting in order to accomplish its goal. To identify different types of forged Media Content, the system uses a mix of spatially and frequency dominated data as the input. The accuracy of the system design is achieved through use of Convolutional Neural Network based Deep Learning Networks for Feature Extraction and Transformer based encoders for classification of input images and video. The system has been designed to provide an optimal balance between Performance, Scalability and Real-time Ability, allowing it to work on either Workstation or Edge Deployment Systems. The proposed system design provides an overall comprehensive framework for the Detection of DeepFakes.

3.1 Functional Requirements

The Deepfake Detection System is responsible for managing the entire detection process, starting with any image or video frame provided by a user. The Deepfake Detection System assesses all input images and video frames in regard to their file formats, quality of content, and integrity. Once verified, these images and/or video frames are processed. The first step of processing involves Preprocessing, which is performed using standard processes by means of resizing the images/video frames to a consistent specification standard and performing normalization or color correction.

Standardized preprocessing methods such as resizing, normalization and color correction are used, providing consistency among the extracted advanced features, which are divided into two main areas:

- **Spatial Branch:** Extracts both pixel-level and texture-based features by using lightweight CNN models that process RGB images.
- **Frequency Branch:** Calculates frequency-domain features by Fast Fourier Transform (FFT) or wavelet-based decomposition.

These characteristics are combined and transmitted to Transformer encoder blocks to sequence model and classify into real and fake. The system also includes:

- Logging, auditing, and secure storage of detection results.
- Visualization dashboards for analytical insights and performance monitoring.
- Optional tagging and categorization of detected manipulation types.

3.2 Non-Functional Requirements

The non-functional system objectives are focused on performance, scalability and reliability:

- **Accuracy:** The system must produce an AUC higher than 0.90 on benchmark datasets.
- **Speed:** CPU-based systems should take less than one second per frame to infer.
- **Portability:** Models must be exportable to TensorFlow Lite (TFLite) or ONNX to run on mobile or edge devices.
- **Robustness:**This requirement is that the detection accuracy should not suffer because of visual distortions like compression, blur, and noise.
- **Security:** Uploaded files, logs, and analytics data should be stored and transmitted encryptively.
- **Scalability:** he architecture is supposed to be scalable, both to batch processing and to multi-user mode so that the analysis of the off-line and large-scale data can be performed.

3.3 Hardware Requirements

The following hardware configuration supports training, evaluation, and deployment of the deepfake detection model.

System Overview: Intel Xeon W-2295 Workstation

- **Processor:** Intel Xeon W-2295, 18 cores / 36 threads, 3.0 GHz base, 4.8 GHz turbo, 24.75 MB cache.
- **Memory:** 128 GB DDR4 RAM (quad-channel, ECC supported), which is suitable to train large models.

- **Operating System:** 64-bit operating system, x64-based processor (Windows 11 Pro, Version 22H2)
- **PCIe Support:** Up to 48 lane support to install cards and accelerate.
- **Performance Features:**
  - Intel Deep Learning Boost and AVX-512 extensions to make AI computation faster.
  - ECC memory to correct errors as well as to be stable in the course of protracted training.
  - Thermal management to maximize continuous operation at workload.
- **Application Suitability:** Best suited to deepfake detection workloads based on CNN-Transformer hybrid models, data augmentation and large-scale inference. The system is compatible or surpasses the recommended hardware of EfficientNet and transformer-based model deployment.

3.4 Software Requirements

his section describes the software requirements of the system's design.

- **Programming Language:** Python 3.8 or above
- **Deep Learning Framework:** PyTorch or TensorFlow/Keras
- **Computer Vision Libraries:** OpenCV, Albumentations, ImgAug
- **Signal Processing Libraries:** NumPy, SciPy (FFT), PyWavelets
- **Development Environments:** Visual Studio Code, Jupyter Notebook
- **Visualization Tools:** TensorBoard, Weights & Biases
- **Supported Operating Systems:** Ubuntu 22.04 LTS, Windows 11 Pro.

3.5 Cost Estimation

The significant cost elements of the project are:

- Cloud/GPU compute resources to model train and experiment with. or cloud volume storage with high capacity to manage the datasets. purchase (CPU/GPU/edge devices to be deployed

- safeguarding of secure storage and analytics infrastructure.

Open-source architectures are used to reduce the cost of licenses. Additional features can consist of GPU servers in order to train faster, mobile hardware in order to test edges, and collaboration platforms to work in a multi-developer environment.

Chapter 4 Methodology

4.1 Tools and Techniques

The proposed approach has been developed using well-established deepfake detection frameworks, and utilises a multi-stage structure that is built upon principles of: transparency, reproducibility, and resiliency to operate in real-world conditions.

The FusedNPZDataset Dataset Preparation is designed to efficiently fuse features into a single .npz file by utilizing EfficientNet embeddings and corresponding frequency domain (RGB) images. This organization results in enhanced loading performance for models trained on this type of optimized data.

To facilitate concurrent calculations on the EfficientNet-B4 model, one of the two backbones utilized within the framework is focused on processing RGB images, while the other backbone is focused on processing frequency domain images. Because both the RGB and frequency domain image backbones are capable of efficiently processing frequency components of spatial data, they form a dual pathway architecture for generating spatial and spectral features concurrently. Both a Fast Fourier Transform (FFT), and a Wavelet Transform have been included to accurately model low frequency distortions associated with the synthetic production of facial images, thus enhancing model sensitivity to the embedded deepfake manipulation artifacts.

Embedding vectors generated through the process of feature extraction in separate frequency and spatial domains are then fused together to create a combined embedding vector that is fed into Transformer Encoder Blocks. The increased computational power of the Transformer Encoder Block also enables it to learn long-term dependencies and contextual relationships among facial components and across multiple video frames. To increase model generalization, two additional enhancements are also included to reduce dependence on large labelled datasets: Self-Supervised Augmentation, and Adversarial Sample Generation.

The model performance has been assessed by both ROC analysis and confusion matrix evaluation, which provide threshold-independent measures and also allow class-wise interpretation of the model. The entire system was developed in Python using the PyTorch library, with support from open-source libraries for data augmentation, visualisation, and numerical computation enabling scalability, flexibility, and reproducibility

when used in various experimental conditions.

4.2 Dataset Description

To perform model training and validation, the research utilized the Celeb-DF dataset as a primary dataset and then applied the FaceForensics++ dataset to test generalisation and robustness across different datasets. The combined datasets also provide an opportunity for exposure to high-quality manipulated media with realism while evaluating results across many different techniques used for synthesising media.

- **Celeb-DF**: The primary dataset for training purposes is made up of high-quality deepfakes made from celebrity images, providing accurate depictions without the visual artifacts typically found in such content. Because of these characteristics, the dataset is ideal for training models that will detect very small manipulation cues in a more-developed version of reality.
- **FaceForensics++**: This dataset can be used to evaluate whether cross-domain transfer learning is effective for generalizing across different offline contexts and performance testing of the different types of manipulation methods to see how well they share information when re-creating a specific face.

Face detection and alignment based on the detection of facial landmarks are included in the preprocessing; resizing to 380×380 pixels and controlled distortions, including Gaussian noise, JPEG compression, and blur augmentation, were added to simulate 'real-world' conditions of recording.

The data sets used to extend the evaluation of the considered models are DFDC and WildDeepfake, as well as additional evaluation will allow for further assessment of adaptive capabilities to unseen variation in manipulation styles and production methods.

Stringent validation criteria for data integrity, traceability, and adherence to ethical guidelines for research on the creation of deepfakes have been applied throughout the data processing pipeline to remain consistent with the ethical recommendations for deepfake research. Metadata-complete samples were kept, whereas samples that failed the auto-validation process were automatically removed from the processing pipeline.

4.3 Flowchart of Methodology

The process for creating a deepfake detection system involves several stages, starting with acquiring input, then progressing through preprocessing of the input, extraction of features, combining these features, classifying them and finally determining how well the classification works.

The first stage of the workflow involves ensuring that all data is valid and that it can be processed properly, as well as performing preprocessing on all data to bring all images to a common size, to normalise all images, to add fake noise, blur, and compression to each of the images so that they reflect the same types of distortions caused by surveillance systems operating on the web.

The feature extraction uses a two-pronged approach, with one set of CNN-based spatial feature branches capturing information about the texture, edge-level details and structural features of a face, and another set of features from the Fourier Transform and Wavelet Transforms where the Wavelet Transform identifies spectrum anomalies that occur as a result of applying synthetic edits to an image. All of the features from the two feature branches are then combined into one unified feature representation.

The fused feature embeddings pass through the Transformer encoder layer to capture the contextual relationships and long-distance dependencies across facial features. This architecture allows the system to recognise subtle manipulation that typically escapes the structural architecture of previous convolutional architectures. Finally, the system makes a prediction about whether the input file is a real or fake file, records and visualises the prediction and uses both the recording and graph to evaluate and compare the performance of the previous work on deepfake detection.

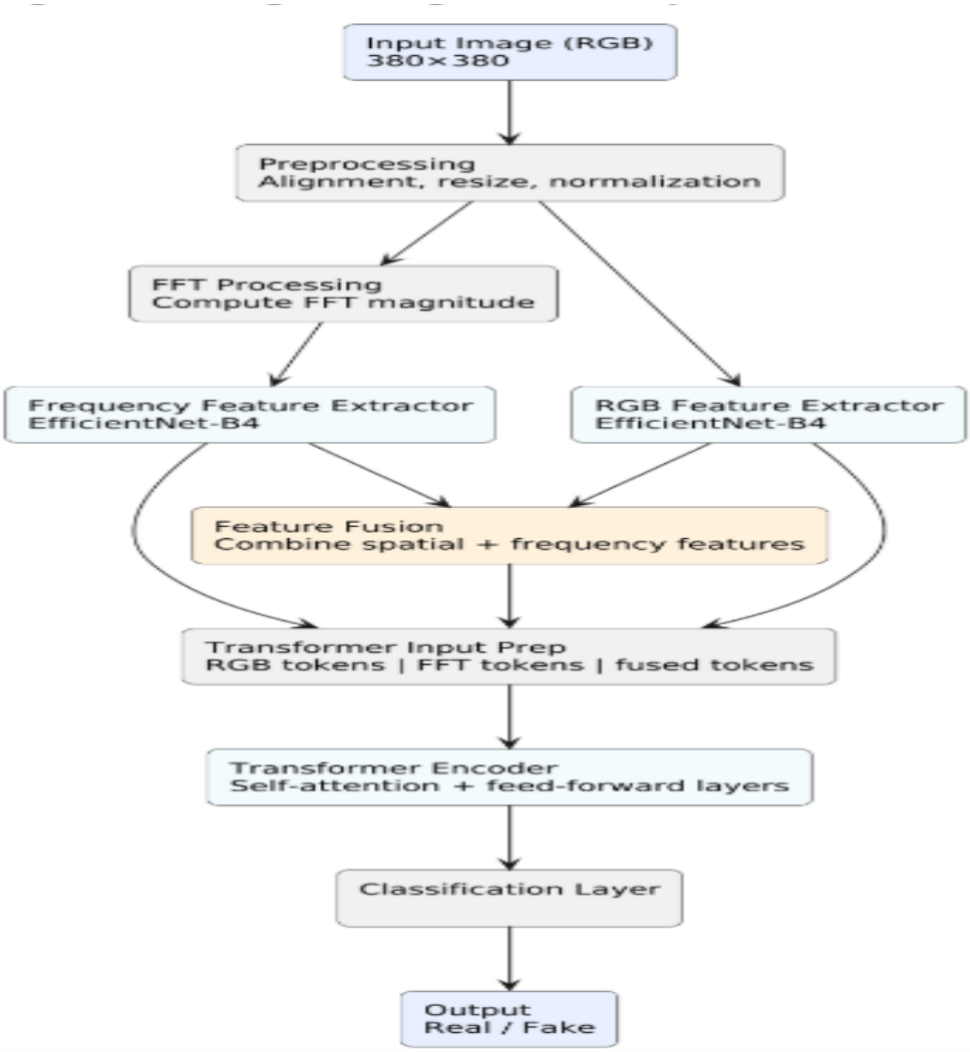

Figure 4.1: Workflow of the Proposed Deepfake Detection Framework

Each stage in the workflow significantly contributes to the overall robustness and reliability of the proposed system:

- **Input Validation:** Ensures data quality, correct formatting, and completeness before processing.
- **Preprocessing:** Includes resizing, normalization, and augmentations such as Gaussian noise, blur, and compression to simulate real-world conditions.
- **Feature Extraction:** Utilizes a dual-domain learning strategy to extract spatial (RGB) and spectral (FFT/Wavelet) features.
- **Transformer Classification:** Employs a multi-head self-attention mechanism to learn global dependencies and generate accurate deepfake predictions.
- **Result Logging:** Stores predictions and visual outputs for interpretability, analysis, and performance monitoring.

4.4 Explanation of Methodology in Detail with Equations and Formulae

A Python-based pipeline loads and processes the dataset and assigns the samples to memory and is then fed through the model to be predicted and analyzed. The purpose of this preprocessing step is to make sure that all the samples have identical structure and compatibility during the training and testing processes.

All the input samples have three fundamental elements

4.4.1 Dataset Loading and Preprocessing

The dataset is loaded and processed using a Python-based pipeline, which allocates samples into memory before being passed through the model for prediction and analysis. This preprocessing step ensures that all samples maintain uniform structure and compatibility throughout training and testing phases.

Each input sample consists of three core components:

- **RGB Image:** Represents the spatial domain and contains visual texture and structural information.
- **Frequency Image:** Computed using Fast Fourier Transform (FFT) or Wavelet Transform to reveal spectral anomalies introduced during manipulation.
- **Feature Tensor:** Precomputed feature embedding stored in `.npz` format, represented dimensionally as $(512 \times 7 \times 7)$.

To ensure data compatibility, and to avoid training interruptions, the data loader has a default mechanism of replacing corrupted, missing, or unreadable files with blank tensors of the appropriate shape. This makes the behavior deterministic during the training process.

Formally, the complete dataset can be represented as:

$$D = \left\{ \left(X_{rgb}^{(i)}, X_{freq}^{(i)}, F^{(i)}, y^{(i)} \right) \mid i = 1, 2, \ldots, N \right\}$$

with $y^{(i)} = 0$ denoting real and $y^{(i)} = 1$ denoting fake.

—

$$D = \{(x_i^{rgb}, x_i^{freq}, y_i) \mid i = 1, 2, ..., N\}$$

where x_i^{rgb} and x_i^{freq} denote RGB and frequency images respectively, and $y_i \in \{0, 1\}$ represents the real/fake label.

4.4.2 CNN-Based Feature Extraction

This paper uses CNN-based feature extraction method to enhance the recognition and identification of the proposed deepfake detection system. It involves two EfficientNet-B4 models, one of which is pre-trained with ImageNet using the RGB images and the other model fine-tuned to accept frequency based input. This two-domain structure allows identifying both spatial and spectral characteristics that can be manipulated to detect manipulation.

Convolutional layers are hierarchical features learning of the CNN that have the following definition:

$$f^{(l)} = \sigma\left(W^{(l)} * f^{(l-1)} + b^{(l)}\right)$$

where:

- $f^{(l)}$: output feature map at layer l,
- $W^{(l)}, b^{(l)}$: learnable kernels and bias terms,
- $*$: convolution operation,
- σ: non-linear activation function (Swish in EfficientNet).

The final CNN representations for RGB and frequency branches are computed as:

$$z_{rgb} = f_{cnn}(x^{rgb}), \quad z_{freq} = f_{cnn}(x^{freq})$$

Each output is flattened to produce a feature tensor of dimension $512 \times 7 \times 7$.

4.4.3 Feature Fusion

To integrate complementary information from both modalities, the embeddings are fused using a learnable weighted formulation:

$$z_{fused} = \alpha \cdot z_{rgb} + (1 - \alpha) \cdot z_{freq}$$

where $\alpha \in [0, 1]$ controls the contribution of each feature source. This weighted fusion enhances robustness by leveraging both spectral and texture cues that are often altered in deepfake content.

4.4.4 Transformer Encoder for Sequence Modeling

The fused representation z_{fused} is reshaped into a token sequence and passed through a Transformer encoder. Each block contains a multi-head self-attention mechanism followed by a feed-forward network.

The self-attention function is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q, K, and V denote query, key, and value matrices derived from the fused features.

The operations per encoder layer are:

$$H^{(l)} = \text{LayerNorm}\left(Z^{(l-1)} + \text{MultiHead}(Z^{(l-1)})\right)$$

$$Z^{(l)} = \text{LayerNorm}\left(H^{(l)} + \text{FFN}(H^{(l)})\right)$$

with the feed-forward network defined as:

$$\text{FFN}(x) = \text{GELU}(xW_1 + b_1)W_2 + b_2$$

4.4.5 Classification and Loss Function

The final output token from the last Transformer layer (CLS token) is input into a linear classifier to produce a binary prediction:

$$\hat{y} = \text{softmax}\left(W_c Z^{(L)} + b_c\right)$$

The model is optimized using binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right]$$

4.4.6 Evaluation Metrics

The trained model is evaluated using multiple performance metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{AUC} = \int_0^1 TPR(FPR^{-1}(x))\,dx$$

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN}$$

These metrics ensure analysis under diverse thresholds and perturbation settings.

4.4.7 Implementation Environment

- **Framework:** PyTorch
- **Visualization Tools:** TensorBoard, Matplotlib
- **Hardware:** NVIDIA RTX A5000 GPU, 128GB DDR4 RAM

This modular methodology ensures scalability, interpretability, and deployment readiness across varying datasets and manipulation conditions, supporting highly accurate and efficient deepfake detection in real-world environments.

Chapter 5 Results and Discussion

In this chapter, the results of the experiment of applying the proposed Deepfake Detection framework to various test samples are provided. The experiments intended to test the ability of the model to withstand, be accurate and generalise in to different conditions. In the evaluation process, spatial (RGB) and frequency-domain features were preprocessed and combination of both features were fused to generate a single representation which was then fed through a transformer-based encoder. This architecture enabled the system to not only analyse visual information but also to obtain contextual relationships in the feature sequences.

5.1 Performance Evaluation

Figure 5.1 and Figure 5.2 test the performance of the model on two benchmark test sets. Their findings demonstrate that the framework is always precise, accurate, and recalls as well as F1-scores, which is an indicator of strong generalisation and not mere memorisation of the patterns of training.

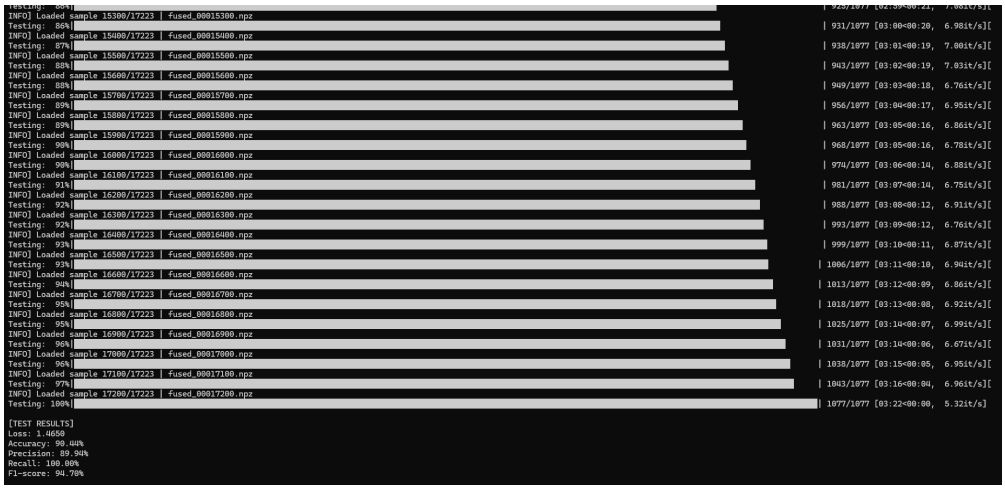

Figure 5.1: **Model Testing on Dataset 1.** Accuracy: 99.01, Precision: 98.14, Recall: 99.92, F1-score: 99.02.

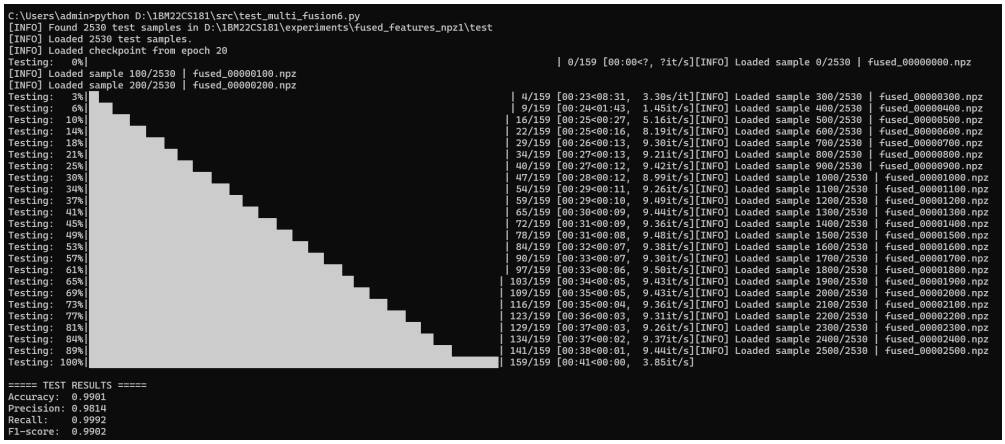

Figure 5.2: **Testing on Extended Dataset.** Accuracy: 90.44, Precision: 89.94, Recall: 100, F1-score: 94.70.

The long dataset has greater diversity and noise than the test set, and this in itself increases the difficulty of the classification problem. Despite that, the model still retains a perfect recall, that is, it is always able to detect manipulated samples in more variable conditions. This minor decrease in accuracy indicates the greater complexity of the data as the model is sometimes misled by various artifacts and real-life bias, which can lead to a false classification of real samples. However, the general performances are good and steady.

5.2 Cross-Dataset Validation

In a bid to explore further the resilience of the model, further tests were conducted using datasets which had varying features including change in compression level, lighting and quality of capture.The results of the detector remain in a high level even in the presence of the distribution shifts. Figure 5.3

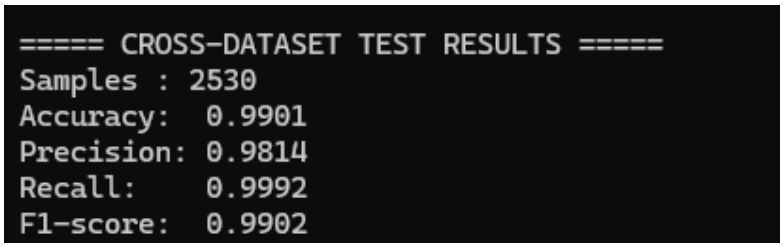

Figure 5.3: **Cross-Dataset Results.** Accuracy: 99.01, Precision: 98.14, Recall: 99.92, F1-score: 99.02.

This stability is a good sign of real-world application, in which input data are usually highly varied and may not reflect the same conditions as during training. The potential

to retain a high level of accuracy and recall when using heterogeneous datasets underscores the generalisability of the suggested fused-feature transformer mode

5.3 Discussion

The experimental findings further validate that the performance of any framework or system that combines spatial and frequency information is much improved for detecting Deepfake video files. The hybrid architecture of the proposed framework allows for very detailed extraction of low-resolution local artefacts or features (i.e., image quality) that are apparent as lack of texture and/or inconsistent regions in the face. In addition, the transformer encoder layer that is fed the output of the encoder provides a mechanism for higher-level feature dependencies. The consistently high-performing F1 scores of the results demonstrate that the proposed framework is capable of maintaining an equilibrium of accurate identification of fake/altered content (recall) while minimizing false alarms/cross-validation errors (precision). For the sake of risk/cost analysis, having both types of errors occur in a real-world scenario will have substantial ramifications. The ability of the proposed framework to perform well across multiple datasets demonstrates the versatility of the architecture. In contrast to many other models that have been designed specifically to work under controlled conditions, the proposed framework has demonstrated the ability to handle different variations of environmental conditions, compression levels, and/or quality of visual forms. The significant findings of this research demonstrate that the proposed Deepfake detection method and architecture will provide a solid base for building more robust and practical detection systems.

Chapter 6 Conclusion and Future Work

6.1 Conclusion

The new deepfake detection model exploits spatial, frequency, and transformer based representations in constructing a robust, efficient and adaptive detection system. A combination of CNNs to extract spatial features with Fast Fourier Transform (FFT) and wavelet transforms to extract information in the frequency domain offers a way of capturing visual discrepancies and the unobtrusive spectral changes as a result of manipulation. The framework with the transformer encoders at the system end point yields high contextual cognition and, thus, can better categorise deepfake videos correctly, hence resulting in higher generalisation of multiple benchmark datasets, including FaceForensics++, Celeb-DF, DFDC, and WildDeepfake.

Experimental analysis demonstrates that hybrid, frequency-attentive frameworks will outperform traditional CNN-only frameworks when video compression, noise, or shifting domains occur in the context of detecting deepfakes. The framework will provide strong potential for deployment in practical applications, enabling accurate detection of deepfakes with speed and computational viability. Collectively, the research presented in this paper contributes to the ongoing development of explainable and cross-domain deepfake detection frameworks, and lays the groundwork for the creation of secure and trustworthy AI-generated media verification frameworks.

6.2 Future Work

We plan to improve and deepen the current system in three interdependent areas of development, namely extending the coverage to more modalities, ensuring the output to be more comprehensible to the final users, and devising the means of successful implementation to devices with limited resources. Certain activities that we will perform include:

1. **Multi-modal fusion:** Expanding the pipeline will include data from both audio (including voiceprints) and text (including text semantics) sources as well as facial images. The combination of signals from multiple modalities will enable

the detection of coordinated manipulation of signals that cannot be detected by single-modality detection methods.

2. **Interpretable decisions:** Analysts can better understand how to develop models for the spatial and spectral data by being able to identify the influx of features in their decisions. With the inclusion of explainable tools such as saliency maps, token-level attention visualization, and counterfactual examples, the models being developed will create greater user trust in their models, and enable them to better identify which mode of failure is created during model development.

3. **Edge-friendly models:** Develop light weight versions of the mobile and embedded versions of the networks using pruning, quantization and using knowledge distillation to reduce the size of networks required for use on Midrange phones and Edge TPUs. Evaluate latency vs. accuracy tradeoffs on selected representative target devices.

4. **Robustness against adaptive attacks:**Increasing augmentations by using artificially generated perturbations from an adversarial manner, and also by incorporating a wider variety of augmentations (for example, simulated compression rates, motion blur, and occlusions) would improve the quality of training. Additionally, the methodology for continually evaluating the detector to adapt to new forgery techniques with less overall training is important.

5. **Secure provenance and forensics:** Investigate a tamper-evident logging system — e.g., anchoring high-level detection metadata to a blockchain or append-only ledger — to support reproducible forensic audit trails and provenance verification.

Together, these efforts aim to transform the prototype into a production-ready toolset that is transparent, resilient to evolving threats, and practical to run on resource-limited devices.

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I Appendix A: Snapshots

Figure 1: User Interface Overview

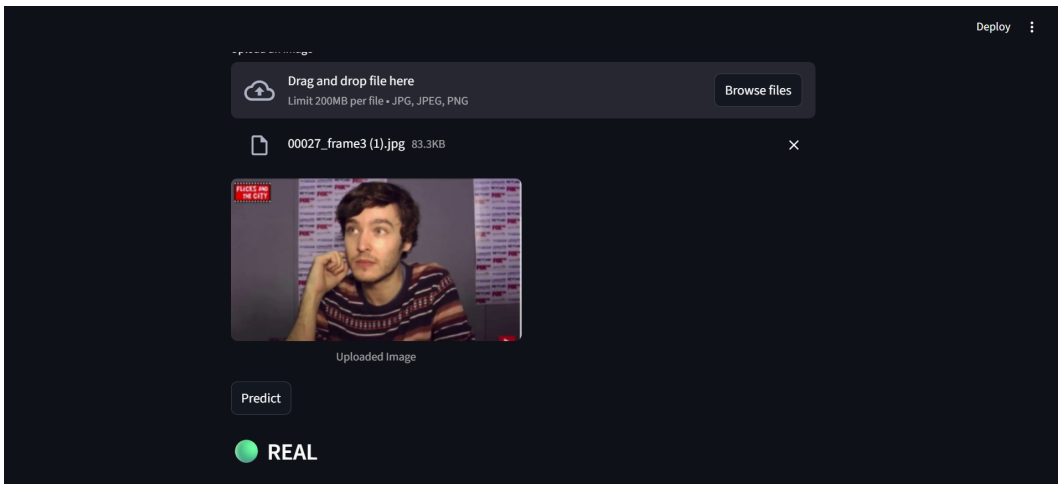


Figure 2: Real Image Upload Screen

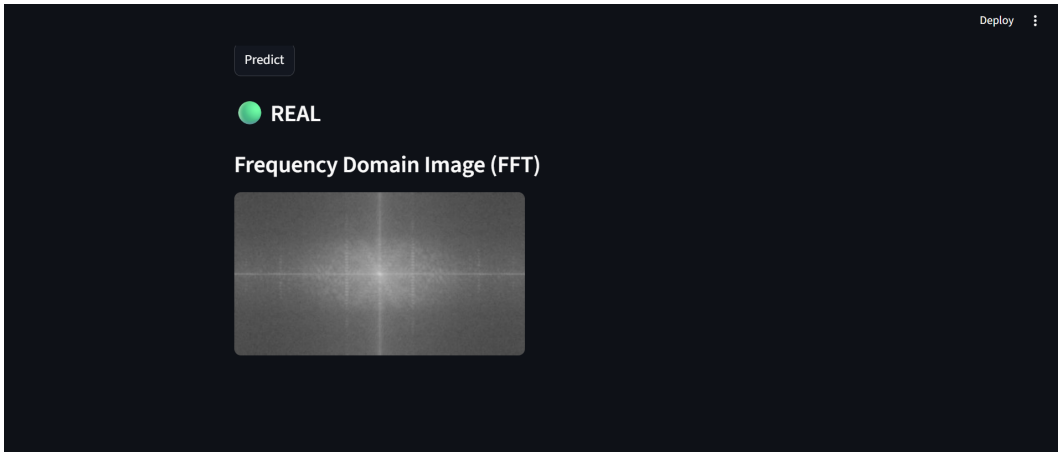


Figure 3: Real Image Detection Result

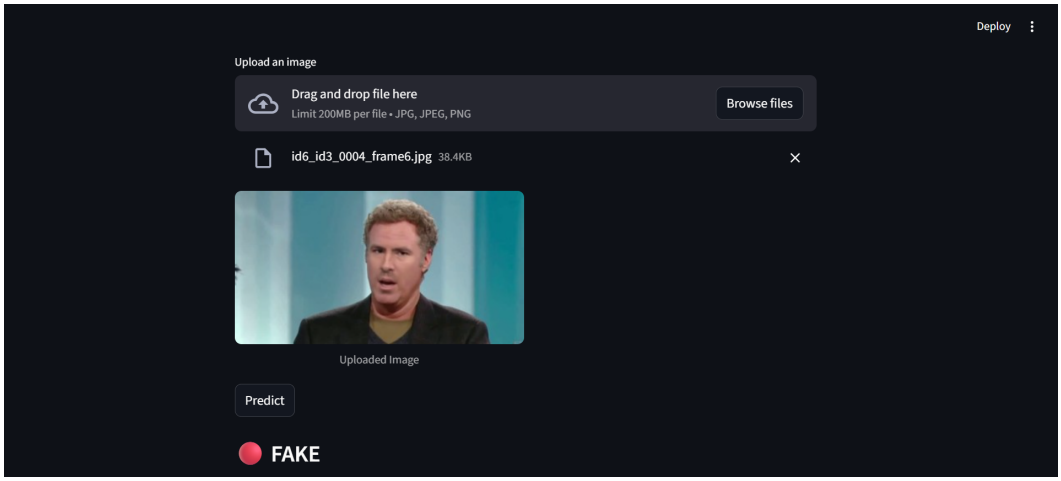


Figure 4: Fake Image Upload Screen

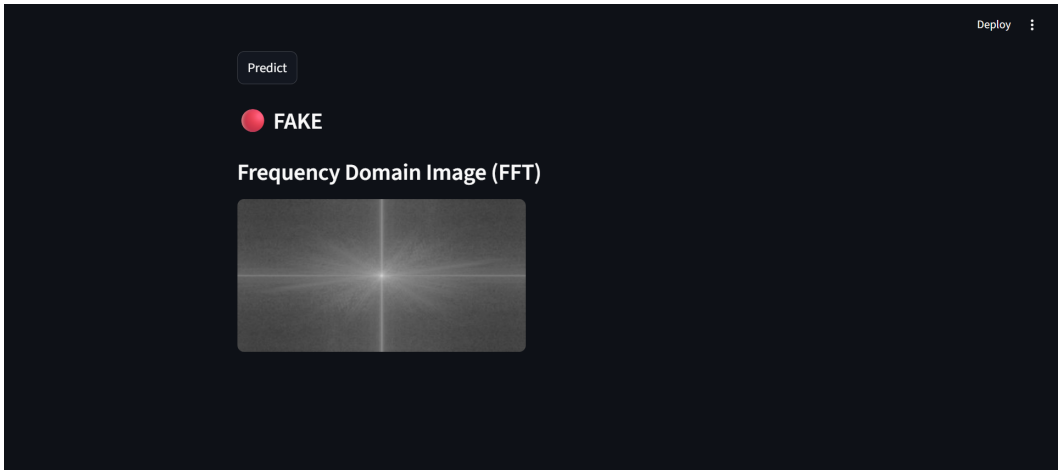


Figure 5: Fake Image Detection Result

II Appendix B: Details of List of Publications Related to This Project

Publication 1

Author Names: Navanith Krishna R, Navya Billalar, Nikhilesh C, Pooja Gaikwad
Paper Title: A Survey on Hybrid Deep Learning Approaches for Robust Deepfake Detection
Name of the Conference or Journal: Academic Survey Paper
Place of the Conference: Bengaluru, India
Date of Conference: 2024

Publication 2

Author Names: Navanith Krishna R, Navya Billalar, Nikhilesh C, Pooja Gaikwad
Paper Title: An Image-Based Deepfake Detection Framework with Dual-Domain CNN Features and Transformers
Name of the Conference or Journal: International Conference on Computer Science and Engineering
Place of the Conference: Bengaluru, India
Date of Conference: 2024

III Appendix C: POs and PSOs Mapped

PROGRAMME OUTCOMES	Level (1/2/3)	Justification if addressed
PO1	3	The project applies engineering knowledge, including AI and ML principles, for detecting deepfake content.
PO2	3	Analyzes the problem of distinguishing real vs fake images using advanced research methods.
PO3	3	Designs a complete deepfake detection solution integrating CNN, Transformers, and dual-domain processing.
PO4	3	Conducts extensive experimentation using datasets such as Celeb-DF and FF++.
PO5	3	Uses modern tools such as Python, TensorFlow, FFT, and Transformer architectures.
PO6	2	Addresses societal needs by mitigating misinformation and promoting digital safety.
PO7	1	Contributes indirectly to sustainable technology by ensuring trustworthy digital media.
PO8	3	Promotes ethical AI use by ensuring transparency and authenticity in digital content.
PO9	3	Involves collaborative work among four authors ensuring teamwork and project organization.
PO10	2	Provides research papers, documentation, and structured outputs for academic usage.
PO11	2	Uses compute resources efficiently and optimizes training for real-world deployment.
PO12	2	Encourages continuous improvement of models through iterative experimentation.

Table 1: Programme Outcomes Mapping with Project

PROGRAMME SPECIFIC OUTCOMES	**Level (1/2/3)**	**Justification if addressed**
PSO1	1	Minimal relevance to networking; uses cloud platforms only for deployment.
PSO2	3	Applies advanced software engineering, ML models, CNNs, and Transformers.
PSO3	3	Integrates AI, signal processing (FFT), and information systems for scalable deepfake detection.

Table 2: Programme Specific Outcomes Mapping with Project

IV Appendix D: Similarity Report and AI-Generated Content Report

This appendix presents the plagiarism similarity report and the AI-generated content verification results for the project document. These reports ensure originality, academic integrity, and compliance with institutional guidelines.

Similarity (Plagiarism) Report

Final_Report_Deepfake (3).pdf

BMS Institute of Technology, Bengaluru

Document Details

Submission ID
trn:oid:::3618:124100747

Submission Date
Dec 9, 2025, 4:31 PM GMT+5:30

Download Date
Dec 9, 2025, 4:36 PM GMT+5:30

File Name
Final_Report_Deepfake (3).pdf

File Size
736.9 KB

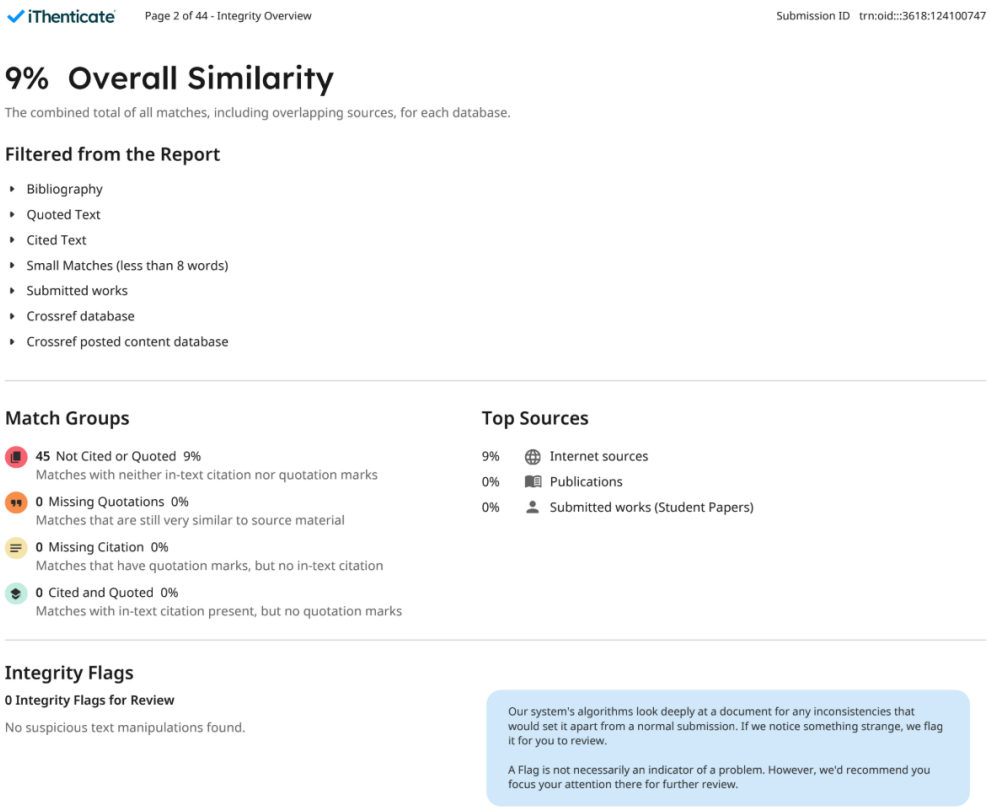
39 Pages

7,861 Words

46,666 Characters

iThenticate Page 1 of 41 - Cover Page Submission ID trn:oid:::3618:124100747

Figure 6: Similarity Report Generated Using the Plagiarism Detection Tool



iThenticate Page 2 of 44 - Integrity Overview Submission ID trn:oid:::3618:124100747

9% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Small Matches (less than 8 words)
- Submitted works
- Crossref database
- Crossref posted content database

Match Groups

- **45** Not Cited or Quoted 9%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

9% Internet sources
0% Publications
0% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Figure 7: AI–Content Evaluation Summary from Similarity Tool

AI-Generated Content Detection Report

iThenticate Page 2 of 41 - AI Writing Overview Submission ID trn:oid:::3618:124100747

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

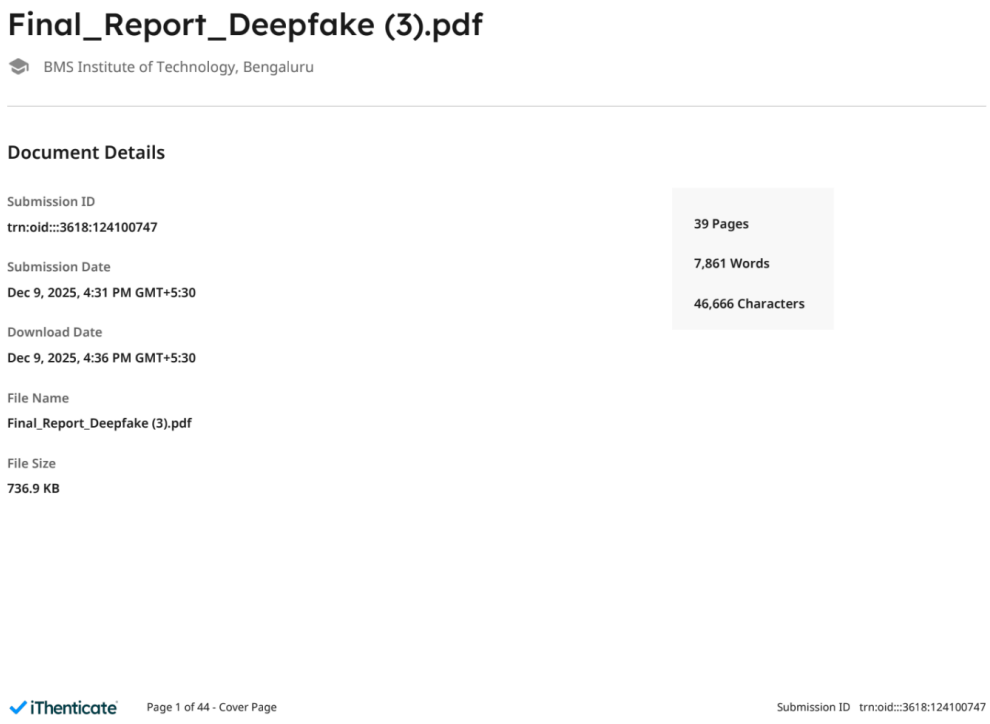
What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Figure 8: AI-Generated Content Analysis Report

V Appendix E: Conference Paper

This appendix includes the screenshots of the conference paper titled **"An Image-Based Deepfake Detection Framework with Dual-Domain CNN Features and Transformers"**. The figures below present the paper as submitted for conference review.



Final_Report_Deepfake (3).pdf

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46,666 Characters

iThenticate Page 1 of 44 - Cover Page Submission ID trn:oid:::3618:124100747

Figure 9: Conference Paper – Page 1

iThenticate Page 2 of 11 - Integrity Overview Submission ID trn:oid:::3618:124090994

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Small Matches (less than 8 words)
- Submitted works
- Crossref database
- Crossref posted content database

Match Groups

25 Not Cited or Quoted 8%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

8% Internet sources
0% Publications
0% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Figure 10: Conference Paper – Page 1 (Continuation)

iThenticate Page 2 of 9 - AI Writing Overview Submission ID trn:oid:::3618:124090994

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

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False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

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Figure 11: Conference Paper – Page 2